Gear Set AGMA

	Input
	Output
	Intermediate Calculations
	Constants
	From a Figure
	Need Drop Down

Material 1144 Carbon Steel

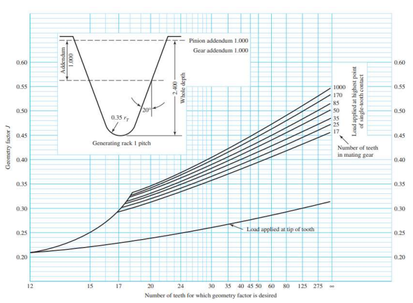
Input Parameters (Pinion)		
Torque	290	lb-in
Speed, n	1700	rpm
Lifetime	97920000	rev
Reliability, R	99	percent (%)
Quality Number	5	
Gear Analyzed dia, d_p	2	in
Mating Gear dia, d_g	6	in
P	10	
F	1	in
Birnell Hardness, Hb	180	
Pressure Angle, X	14.5	degrees
Pressure Angle, X	0.253072742	rad
Safety Factor, S_F		Bending
Safety Factor, S_H		Contact

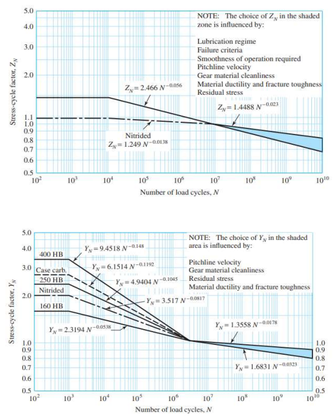
Output Values					
Stress		Unit	Safety Factor		
σ_B	20586.18	psi	S_F	1.205502	
σ_{allow}	24816.68	psi			
σ_C	97415.98	psi	S_H	8.026537	
σ_{allow}	781913	psi			Gear Analyzed (E)
σ_B	81869.56	psi	S_F	3.305451	
σ_{allow}	24768.05	psi			
σ_C	180952.5	psi	S_H	2.366429	
σ_{allow}	76466.47	psi			Mating Gear (D)

Assumtions		
Useage	4	Hours/Day
	2	Days/Week
	24	Weeks/Year
	5	Total Years
Uncrowned Teeth		Cm = 1
Cpm = 1		
Open Gearing		
Ce = 1		
Assumed R=0.99		
Kt = 1		
All steel	Cp = 2300	
Cf = 1		
Ch = 1		

Calculations (Pinion)												
Velocity, V	445.0589593	ft/min										
Power	7.825396825	Hp										
Wt	290	lb										
Ko	1											
Kv	1.038167346		A	504.7697	B	0.914826						
Ks	1	= 1 to avoid calc										
Km	1.167707		Cmc	1	Cpf	0.025	Cpm	1	Cma	0.142707	Ce	1
Kb	2.400824808		a	0.1	h	0.2	t	0.1	mb	0.5		
J	0.41		NA	20	NM	60						
Cp	2300											
Cf	1											
I	0.097986105		mg	3	mn	1						
Kr	1											
Kt	1											
St	26714	Grade 1										
Yn	0.928976598											
Sc	87060	Grade 1										
Zn	0.880043772											
Ch	1											

Input Parameters (Gear)		
Torque	290	lb-in
Speed, n	1700	rpm
Lifetime	97920000	rev
Reliability, R	99	percent (%)
Quality Number	5	
Gear Analyzed dia, d_p	6	in
Mating Gear dia, d_g	2	in
P	10	
F	1	in
Birnell Hardness, Hb	180	
Pressure Angle, X	14.5	degrees
Pressure Angle, X	0.253072742	rad
Safety Factor, S_F		Bending
Safety Factor, S_H		Contact





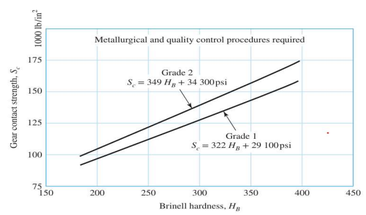
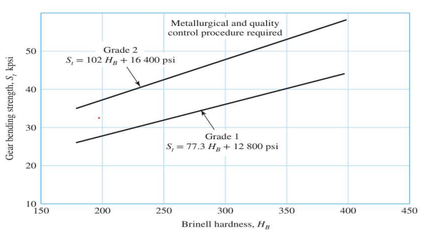


Figure 14–5
Gear contact strength S_c at 10^7 cycles and 0.99 reliability for through-hardened steel gears. The SI equations are: $S_c = 2.22H_B + 200$ MPa, grade 1, and $S_c = 2.41H_B + 237$ MPa, grade 2. *(Source: ANSI/AGMA 2001-D04 and 2101-D04.)*

Figure 14–2
Gear bending strength for through-hardened steels, S_t. The SI equations are: $S_t = 0.533H_B + 88.3$ MPa, grade 1, and $S_t = 0.703H_B + 113$ MPa, grade 2. *(Source: ANSI/AGMA 2001-D04 and 2101-D04.)*



Calculations (Pinion)												
Velocity, V	2670.353756	ft/min										
Power	7.825396825	Hp										
Wt	96.66666667	lb										
Ko	1											
Kv	1.082966407		A	509.773	B	0.825482						
Ks	1	= 1 to avoid calc										
Km	10.7494		Cmc	1	Cpf	8.975	Cpm	1	Cma	0.7744	Ce	1
Kb	2.400824808		a	0.1	h	0.2	t	0.1	mb	0.5		
J	0.33		NA	60	NM	20						
Cp	2300											
Cf	1											
I	0.030300601		mg	0.333333	mn	1						

Kr	1.00196355	Assumed R=0.99										
Kt	1											
St	26714	Grade 1										
Yn	0.928976598											
Sc	87060	Grade 1										
Zn	0.880043772											
Ch	1											